# EXPERIMENT – 3

**AIM :**

To write a Python program to solve the Water Jug Problem by measuring a required quantity of water using two jugs of different capacities.

**ALGORITHM:**

1. Start.
2. Initialize two jugs with their capacities.
3. Set the initial water level of both jugs to zero.
4. Define the possible operations: fill, empty, and pour water between jugs.
5. Apply the operations to reach the required quantity.
6. Repeat until the goal state is reached.
7. Display the final water levels.
8. Stop.

**PROGRAM :**

Experiment 3.py - D:/Abhi/Collage/AI/AI-Lab experiment-2026/Experiment 3.py (3.14.3)

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```
from collections import deque

def water_jug(cap1, cap2, target):
    queue = deque([(0, 0, [])])
    visited = set()

    while queue:
        a, b, path = queue.popleft()

        if (a, b) in visited:
            continue
        visited.add((a, b))
        path = path + [(a, b)]

        if a == target or b == target:
            return path

        states = [
            (cap1, b), (a, cap2), (0, b), (a, 0),
            (a - min(a, cap2 - b), b + min(a, cap2 - b)),
            (a + min(b, cap1 - a), b - min(b, cap1 - a))
        ]

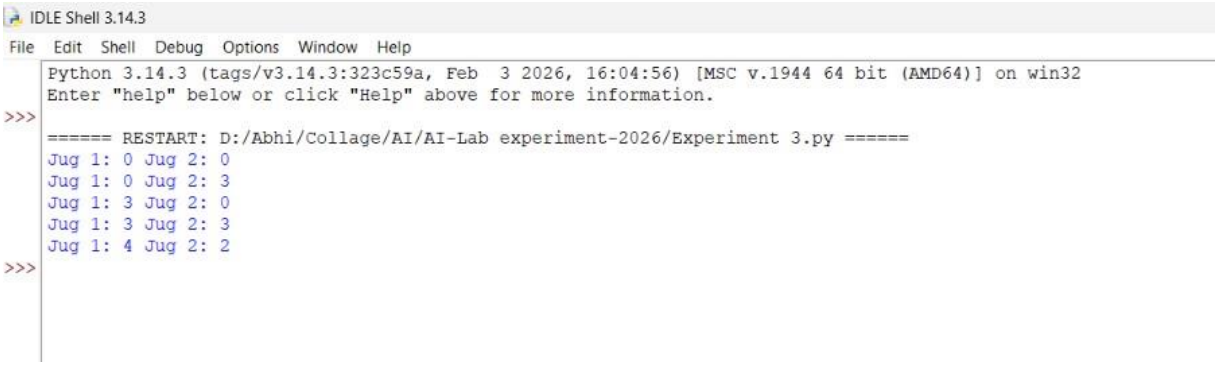
        for state in states:
            if state not in visited:
                queue.append((state[0], state[1], path))

    return None

result = water_jug(4, 3, 2)

if result:
    for state in result:
        print("Jug 1:", state[0], "Jug 2:", state[1])
else:
    print("No solution")
```

**OUTPUT:**




**RESULT:**

Thus, the Python program to solve the Water Jug Problem was executed successfully, and the required quantity of water was obtained.